

SAPIENS APP — MVP BUILD SPECIFICATION

HOW TO USE THIS

Build in the phase order of §7. Each phase is a working milestone. Do ONE phase at a time; stop, let the owner test, then continue. Do **not** build everything at once.

This spec is P1 only. Anything tagged **[P2]** or **[LATER]** in the PRD must **not** be built now but the architecture must leave room for it (§8). The explicit "do not build" list is §8.

When behaviour detail is needed, read the referenced PRD section (e.g. "PRD 0.4"). Don't re-derive product decisions once they're settled.

The constitution constrains the code (PRD §0): no money for help; no profile surfing (information is earned enforced in data access, not just UI); everyone verified; trust over engagement; never endanger the person being protected; no attention-farming (the anti-addiction rules, PRD 8.9, are hard).

The owner is a solo builder. Explain as you go; prefer clear, maintainable code over clever code.

1. STACK & SERVICES

Layer	Choice
App	React Native + Expo (TypeScript), iOS + Android, one codebase
Backend	Supabase — Postgres · Realtime · Auth · Storage · RLS · Edge Functions (Deno) · PostGIS
Dispatch timing	pg_cron / scheduled Edge Functions (approximate ~1-min tick; graduate to a worker later)
Live delivery	Supabase Realtime (open app) + Expo Push (closed app) — pings fire both
Location	Expo Location (free GPS) + deep-link to external maps (no Map API in v1)
Media	Supabase Storage
SMS	provider module — MSG91 / SmartPing / Twilio (DLT-registered)
Admin / Trust & Safety	Next.js dashboard on Supabase , Vercel-hosted (mirrors the website admin)
Analytics	PostHog
Push	Expo Push

Dependencies (app): expo, expo-router, expo-location, expo-notifications, expo-image-picker, expo-av (voice notes), expo-secure-store, @supabase/supabase-js, react-native-maps is **NOT** used in v1 (external deep-link instead), posthog-react-native, zod. **Dependencies (admin):** next, @supabase/supabase-js, tailwindcss.

Env vars:

app (.env / eas secrets)

EXPO_PUBLIC_SUPABASE_URL=

EXPO_PUBLIC_SUPABASE_PUBLISHABLE_KEY= # new-format key (like the website's anon slot)

EXPO_PUBLIC_POSTHOG_KEY=

server (Edge Functions / admin)

SUPABASE_SERVICE_KEY= # new-format secret key — server only

KYC_PROVIDER_KEY= # chosen provider

SMS_PROVIDER_KEY= # MSG91/etc

SMS_DLT_TEMPLATE_IDS= # DLT-registered template refs

EXPO_ACCESS_TOKEN= # for sending push from server

2. PROJECT STRUCTURE

A monorepo with three parts:

/app → Expo (React Native) — the mobile app

/admin → Next.js — the Trust & Safety dashboard

/supabase → migrations (SQL schema + RLS), Edge Functions, seed data

The app uses **expo-router** (file-based). Shared types (DB row types) generated from Supabase and imported by both /app and /admin.

Design language inherits the website (PRD 10.2): warm paper/ink/spark + celestial; **Nunito Sans** for all UI, **Cabin Sketch** for celebratory moments only. Build a small themed component kit first (Button, Card, Tile, Screen, Sheet) so screens are consistent and the 70-year-old test (PRD 10.1) is met, large touch targets, high contrast, minimal steps.

3. DATA MODEL (Supabase / Postgres)

Core P1 tables, grouped by domain. Columns are indicative; Claude Code writes the migrations. **RLS intent is stated per table and is not optional, it enforces the constitution.**

Identity & profile

profiles (id = auth uid) · display_name (first name) · display_photo_url · kyc_selfie_ref (private, never exposed) · verified bool · over_18 bool · verification_token · member_since · bio · cached: celestial_stage, unique_helps, total_helps (steadfast), moneta_lifetime, moneta_balance, trust_rating_avg, goodness_score.

RLS/exposure: owner reads all. **Others read ONLY** display_name, display_photo_url, celestial_stage, trust_rating_avg, member_since enforce via a **public view** (profiles_public) exposing just those columns; the full row is owner-only. This is the staged-disclosure boundary (PRD 0.4, 9.4). Connections may read a fuller view (PRD 5.4).

helper_preferences · user_id · categories (uuid[]) · radius_max_m · quiet_hours (jsonb). Availability is implicit — no toggle (PRD 3.1). RLS: owner-only.

trusted_contacts · user_id · name · phone. 1–3 (PRD 1.2). RLS: owner-only. Used by SOS via SMS.

Requests, categories & dispatch (the engine)

- **categories** · id · parent_id · label · slug · enabled · version · icon · default_timing (now/scheduled/either) · typical_urgency · interaction_type (one_to_one/group/either) · implies_physical_meet · safety_sensitivity · allows_media · radius_hint. **Seeded from PRD 3.11.** Editable data. RLS: public read (enabled only); write = service/admin.
- **category_suggestions** · user_id · text · status (pending/approved/rejected). (PRD 9.6) RLS: owner insert; admin read/update.
- **requests** · id · seeker_id · category_id · description · voice_note_url? · timing · scheduled_at? · urgency · status (open/matched/active/completed/cancelled/expired) · meetpoint_lat/lng (**released to helper only at confirm**, PRD 0.5) · approx_area · interaction_type · participant_cap? · is_directed · directed_to? · prefer_women · created_at · expires_at. RLS: seeker owns; helpers see **only approx fields** until confirmed (via a dispatch view).
- **dispatch_targets** · request_id · helper_id · wave_number · pinged_at · outcome (pending/raised/ignored/expired). Drives wave logic + fairness rotation. RLS: helper reads own; service writes.
- **request_responses** (raised hands) · request_id · helper_id · status (raised/confirmed/vetoed/withdrawn). RLS: seeker + the responder.

Matching, meeting, completion

- **matches** · id · request_id · helper_id · seeker_id · status (confirmed/on_the_way/arrived/completed/cancelled) · meetup_code (short) · confirmed_at · helper_done_at · seeker_confirmed_at · auto_confirmed bool · completed_at. Group requests → multiple matches per request. RLS: the two parties only.

- **ratings** · match_id · rater_id · ratee_id · stars (0–5) · feedback_text? · feedback_voice_url? · submitted_at · revealed bool. **Double-blind** (PRD 0.8): neither side reads the other's until both submitted — enforce at API/view level.

Communication

- **chats** · id · request_id (active-request) OR connection_id (inbox) · kind (active/inbox) · created_at · closed_at?. Active-request chat **dissolves at confirmed completion** (PRD 4.2); inbox persists.
- **messages** · chat_id · sender_id · type (text/photo/voice/emoji) · body? · media_url? · created_at. **Active-request messages are server-readable** (evidence, PRD 6.2) and purged after a retention window (PRD 4.3). Inbox = encrypted in transit/at rest, block-based (E2E is [P2]). RLS: chat participants; **admin can read flagged active-request chats** (via service role, audited).

Connections & social graph

- **connections** · id · user_a · user_b · status (offered/active/declined/disconnected) · formed_from_request? · offered_at · active_at? · disconnected_at?. Offer expires 7 days (PRD 5.3). Disconnect = silent, one-sided, history frozen-but-visible (PRD 5.9). RLS: the two parties.
- **blocks** · blocker_id · blocked_id. Mutual invisibility; **dispatch must never match a blocked pair** (PRD 5.10). RLS: owner.

Reward & reputation

- **moneta_ledger** (append-only, immutable) · user_id · help_ref (match id) · amount · type (earned/spent/reversed) · counterparty_id · created_at. **Unique-help rule (PRD 7.1): award only on the FIRST completed help between a given pair** — enforce with a uniqueness check on (helper, seeker) first-completion. RLS: owner read; service write.
 - Derived: **unique_helps**, **moneta_lifetime**, **moneta_balance** (lifetime never decreases; balance = earned – spent). **total_helps** (steadfast) counts all completions incl. repeats (PRD 7.2).
- **Meters (PRD 7.7)** computed, not stored raw: **Trust** = avg stars (0–5); **Goodness** = $100 \times (1 - e^{-(\text{unique_helps} / k)})$ (owner-facing, tune $k = A21$); **Spiritual** = Celestial stage from unique_helps thresholds (PRD 7.8). Goodness does **not** decay.
- **leaderboard** — a query/materialized view: **new lifetime-unique people met within the month**, by zip/city/state/country (PRD 7.9). Auto-opt-in.

Safety

- **reports** · reporter_id · reported_id · context (chat/match/moment) · evidence_chat_id? · reason · status (open/reviewing/actioned/dismissed) · resolution?. Powers the T&S loop (PRD 1.4). RLS: reporter insert; admin read/act.
- **sos_events** · user_id · lat/lng · created_at · resolved (are-you-safe step) · daily_count. **Soft daily limit 2–3** (PRD 10.9): after limit, still fire 112 + trusted-contact SMS, but stop new community/system events and flag. RLS: owner + admin.

Moments, appreciation, notifications

- **moments** · type (selfie/milestone) · request_id? · participants (uuid[]) · photo_url? · caption · area (zip/locality) · consent_a/consent_b bool · visible bool · created_at · removed_by?. **Selfies: double**

opt-in, area-scoped, either party can remove anytime, NOT searchable, no tap-to-profile (PRD 8.4). RLS: area-scoped read; participants can remove.

- **appreciations** · **moment_id** · **user_id**. ❤️ gesture; **count hidden** (PRD 8.7).
 - **notifications** · **user_id** · **type** · **payload** (jsonb) · **read** · **created_at**. Separate budgets per type (PRD 3.8/5.8/6.11).
 - **Activity map** = an **aggregate query only** — helps per area per week; **never an individual pin, never SOS** (PRD 8.6).
-

4. CROSS-CUTTING SYSTEMS

- **Staged disclosure is enforced in DATA, not just UI** (PRD 0.4): the dispatch/list views expose only approx location + limited profile until a **matches** row is **confirmed**; precise **meetpoint** and the navigate deep-link are released only then. Build the views/policies so a client *cannot* fetch what the ladder hasn't unlocked.
 - **The two-photo model** (PRD 9.2): **kyc_selfie_ref** is private and never served. On display-photo set/change, run a **face-match against the KYC selfie** (via the KYC/liveness provider or a face-match API); only on match does it become **display_photo_url**. On fail: keep the previous photo, soft-flag repeated failures to **reports/admin**.
 - **Realtime + Push split** (PRD 11.5): every dispatch ping, match update, and message triggers a Realtime event (open app) **and** an Expo Push (closed app).
 - **SMS module** (PRD 11.7): one wrapper; DLT-registered templates for OTP + SOS. SOS SMS must send even on poor data (PRD 11.8).
 - **Location** (PRD 10.10): Expo Location for GPS; a **LocationProvider** module returns coordinates + distance + ETA; a **showLocation()** abstraction deep-links to external maps now, swappable for an embedded map later. Seeker sees text/ETA ("Raj is 400m away · ~3 min"), not a live dot.
-

5. THE DISPATCH ENGINE (the custom core — build carefully)

This is the one genuinely hard part (PRD 3.x, 11.3). Build and test it with **seeded data** before wiring the UI.

Request lifecycle (state machine): **open** → **matched** → **active** → **completed** (with **cancelled/expired** branches).

On request creation:

1. Resolve eligible helpers via **PostGIS**: within current radius, **categories** $\supseteq$ request category, verified, not mid-help, not in quiet hours, **not blocked** with the seeker, honoring **prefer_women**.
2. **Order**: nearest-first, corrected by **load-rotation fairness** (PRD 3.4) — do not rank by badge/reputation.
3. **Wave dispatch**: ping the first wave (nearest few); write **dispatch_targets**. **Urgency compresses timing** (PRD 3.3): SOS/urgent $\approx$ all-at-once, casual = patient waves.
4. Send ping via **Realtime + Push**; helpers see limited info (PRD 10.6).

Scheduled tick (pg_cron, ~1 min):

- For open requests past their wave interval with no acceptance → **widen radius (adaptive, up to helper max)** and ping the next wave (PRD 3.2/3.3).
- For confirmed matches where the helper hasn't hit "on my way" within the timeout → **lapse and silently re-broadcast** (PRD 0.7).
- For requests past **expires_at** with no match → mark expired, offer the seeker "notify me when someone can help" (PRD 3.10).

Accept models by tier (PRD 0.3 / 3.6):

- **Casual/Everyday** → **first-to-accept**: first raised hand wins; create **matches** (confirmed); seeker sees "X is coming" with a decline/veto option before location/chat unlock.
- **Urgent** → **auto-assign**: system picks the best single helper, asks one at a time; seeker still gets the veto.
- **Seeker veto**: on decline, silently re-offer to the next; the vetoed helper sees only "matched with someone else" (PRD 10.6). **Seeker-picks-from-a-list is [P2] — do not build.**
- **Group (scheduled)** → **multi-accept up to participant_cap**; a group coordination chat (PRD 4.7).

On confirm: release precise **meetpoint**, generate **meetup_code** + navigate deep-link, open the active-request chat, begin ETA status.

SOS = dispatch with the gate removed + widest reach, but in v1 it does **not** page the community (Layer 3 is [P2]) — it fires **trusted-contact SMS + one-tap 112** (PRD 1.2). Same trigger, different config.

6. SMS, KYC & PUSH INTEGRATIONS

- **KYC:** provider SDK/flow at the verification gate (PRD 2.1/2.2) — Aadhaar via DigiLocker/offline-QR + DL fallback + liveness selfie; store only **verified/name/photo/over_18/token** — **never a raw Aadhaar number**. Also used for display-photo face-match (§4).
- **SMS:** OTP (Supabase Auth can drive phone OTP; SMS provider for delivery/DLT) + SOS trusted-contact alerts with a live-location link.
- **Push:** Expo Push tokens stored per device; server sends via **EXPO_ACCESS_TOKEN**.

7. BUILD ORDER (phases — each a testable milestone)

Phase 0 — Foundation. Monorepo; Expo app scaffold + expo-router; Supabase project; the full schema (§3) + RLS + **profiles_public** view; themed component kit; auth (email + phone OTP); env wiring; PostHog. *Milestone: sign up, log in, empty home.*

Phase 1 — Identity & profile. Onboarding flow + short walkthrough + trusted-contacts capture + "This is Sapiens" welcome view (PRD 10.7); KYC provider integration + the gate (PRD 2.1); two-photo model with face-match (PRD 9.2); the "You" tab (PRD 10.11); "Ways I help" + helper preferences + the category suggestion box (PRD 9.6). *Milestone: a fully verified user with a profile and preferences.*

Phase 2 — THE SPINE (the core product). Categories seeded (PRD 3.11); raise-help flow, 3 taps (PRD 10.5); the **dispatch engine** (§5) with seeded-data tests; helper ping + Help-now bounded list (PRD 10.6); match → confirm → staged disclosure release; active-request chat (text/photo/voice/emoji, server-readable, PRD 4.x) with the mid-request **Cancel+Report+Block** (PRD 6.13); location ETA + navigate deep-link (PRD 10.10); meetup code; completion (helper done → seeker confirm/auto-confirm, PRD 0.8). *Milestone: a real help, end to end, between two test users.*

Phase 3 — Reward & reputation. Moneta ledger with the **unique-help rule** (PRD 7.1) + steadfast count (7.2); double-blind ratings (0.8); the three meters (7.7); Celestial Journey visuals + milestones (7.8); the leaderboard view (7.9). *Milestone: helping moves the moon and the meters.*

Phase 4 — Connections & inbox. Post-completion Connect offer (double opt-in, PRD 5.2); connection graph + privacy (5.7); the permanent **Inbox** (6.x) with block; directed requests + "Ask [name] for help" (5.5); connections wave in dispatch (5.6). *Milestone: help becomes a lasting connection.*

Phase 5 — SOS. Guarded SOS button (press-hold/slide, PRD 10.9); trusted-contact SMS alerts + live-location link (Layer 1); one-tap 112 (Layer 2); soft daily limit + accountability + are-you-safe step; **offline degradation** (PRD 11.8). *Milestone: SOS works, honestly, even on poor data.*

Phase 6 — Community & notifications. Finite community-moments surface with the **anti-addiction hard rules** (PRD 8.9 — bottom, no infinite scroll, "you're all caught up"); double-opt-in area-scoped selfies (8.4); milestones in-feed (8.5); ❤️ appreciate, count hidden (8.7); aggregate activity map (8.6); the notifications bell (10.12); warm empty states (10.13). *Milestone: the app feels alive without a scroll trap.*

Phase 7 — Admin / Trust & Safety dashboard. Next.js on Supabase (PRD 11.9): resolve reports, read flagged active-request chats (evidence), ban/suspend, approve category suggestions. *Milestone: the safety model has an operator.*

Phase 8 — Hardening & store prep. Accessibility pass (70-year-old test, PRD 10.1); security review (RLS coverage, audited access, PRD 11.12); notification budgets/caps; data-retention purge jobs (4.3/6.7); performance on a mid-range Android; store assets + the App Store safety-review readiness (PRD 11.10). *Milestone: submittable.*

8. EXPLICITLY OUT OF SCOPE — DO NOT BUILD

[P2] (leave architectural hooks, per the referenced sections, but do not build): Premium Choice / seeker-picks-from-a-list (0.9) · masked calling (0.6) · SOS Layer 3 community responders (1.2) · E2E inbox + client-side reporting (6.2) · retroactive help-logging (5.11) · voice-input category/sentiment analysis (10.5) · the anti-collusion layer (1.5 — safe to defer *only while Moneta has no cash value*, 7.4) · embedded map (10.10) · real-world Moneta redemption (7.3, **gated behind anti-collusion + G7**).

[LATER]: formal background checks · behavioural trust gates · direct DigiLocker integration · cash rewards & City Saviour tooling · Hindi/regional languages (keep strings i18n-ready) · disappearing messages · arbitrary group chats · comments.

9. HUMAN-ONLY TASKS (owner, not Claude Code)

☐ Create service accounts + paste keys: **Supabase**, **Expo/EAS**, **PostHog**, **KYC provider**, **SMS provider**, **Apple Developer** (\$99/yr) + **Google Play** (\$25 once). ☐ **KYC provider needs a registered business entity** — this ties into the legal-entity gate (PRD G1) and may not be fully deferrable. ☐ **DLT registration** for SMS sender IDs + templates (PRD 11.7) — required before OTP/SOS SMS works. ☐ Choose the actual **KYC vendor** and **SMS provider** (Claude can research current options + pricing when you're ready). ☐ **Pre-launch legal gate G1–G7** (PRD 2.4) — engage a lawyer with Indian tech/platform experience. Blocks public launch. ☐ **App Store safety review** (PRD 11.10) — have moderation/reporting/SOS demonstrably working before submission. ☐ **Funding** (PRD 12.3) — its own session; the only truly unfunded piece. ☐ Launch target: **closed communities** (a specific college/company/society) — pick the first one (PRD 12.1).

End of MVP Build Specification. Build P1 in phase order; keep P2/LATER out; read the PRD for behavioural detail. Where unspecified, default to: warm, calm, human, radically simple — and never compromise safety or the constitution for speed.